

Muhammad Tayyab Tanveer

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EDUCATION

COMSATS University Islamabad — Main Campus

2024 – Present

Bachelor of Science in Software Engineering | CGPA: 3.1 / 4.0

Relevant Coursework: Data Structures & Algorithms • Object-Oriented Programming • Database Systems • Software Requirements Engineering • Information Security • E-Commerce & Digital Marketing

INTERNSHIP EXPERIENCE

Machine Learning Intern — Arch Technologies (Remote)

Apr 2026 – May 2026 | 8 Weeks

Machine Learning Internship & Training Program — Pakistan's Digital AI Training Platform

- Completed an 8-week remote Machine Learning Internship and Training Program at Arch Technologies
- Gained hands-on experience with machine learning concepts, model development, and practical AI application under senior mentorship
- Worked under guidance of Sr. Python Developer on Python-based ML workflows
- Awarded Certificate of Completion (Cert ID: 993817) by CEO & Sr. Python Developer upon successful program completion

AI-Powered Data Analysis Intern — Excelerate (Global Remote)

2026 | 4 Weeks

Global Remote Internship Program

- Completed a structured 4-week remote internship focused on AI-driven data analysis methodologies
- Applied analytical tools and techniques to interpret datasets and generate actionable insights
- Awarded Certificate of Completion upon successfully fulfilling all program requirements

ACADEMIC & PERSONAL PROJECTS

PaceTrack — Full-Stack Running Tracker Mobile App

Solo Full-Stack Developer

Full-stack mobile running tracker with live GPS tracking, route planning, and gamified motivation features — deployed end-to-end

- Built a secure authentication system with email OTP verification, bcrypt password/OTP hashing, and JWT session management
- Implemented real-time GPS run tracking using Haversine distance calculation, achieving live pace/distance updates from raw location streams
- Integrated third-party routing (OSRM API) for road-accurate route planning with live polyline rendering on interactive maps
- Designed a MongoDB aggregation pipeline for dashboard analytics (daily/weekly stats, streak calculation, achievement unlocking logic)
- Built a gamification system (streaks, achievement badges) and local push notification engine to drive user engagement
- Implemented voice-guided run coaching using text-to-speech, triggered dynamically at distance milestones
- Designed and built a full light/dark theme system with persisted user preferences
- Deployed a production REST API (Render) connected to MongoDB Atlas, and shipped a distributable Android build via EAS

Tech Stack: React Native (Expo, TypeScript), Node.js, Express, MongoDB Atlas, JWT, React Navigation, react-native-maps, react-native-reanimated

GitHub (Mobile): <https://github.com/tayabawan19/PaceTrack>

GitHub (Backend): <https://github.com/tayabawan19/PaceTrack-backend>

CropSense — Pakistan Agricultural Intelligence Platform Database Systems | MongoDB + Node.js + Express

Full-stack agricultural management system that digitizes farmer, crop, production, expense, and market data for Pakistan

- Designed a 10-collection MongoDB document schema by migrating a full Oracle SQL relational model — demonstrating mastery of NoSQL denormalization and embedded document strategy

- Created 45 indexes across all 10 collections: system, unique, compound, single-field, and multikey (embedded array) indexes — drastically reducing query time by eliminating full collection scans
- Built a custom Index Manager UI tab showing all 45 indexes organized by collection with color-coded type badges and one-click mongosh query copy — a feature rarely seen in student projects
- Used MongoDB aggregation pipelines (\$group, \$sum) on the dashboard to compute total yield per crop, total expenses per category, and farmer/crop counts in real time
- Implemented embedded document design — land parcels inside farms, expenses inside farm_records, prices inside markets — reducing the need for joins and improving read performance
- Built full CRUD operations across 10 entities (Farmers, Farms, Crops, Records, Production, Expenses, Markets, Weather, Districts, Admin) with live search and toast notifications

Food Delivery System

Data Structures | C++

Console-based application implementing core Data Structures and Algorithms concepts

- Implemented Dijkstra's algorithm to compute the fastest delivery route between locations
- Features include: Add/Search/Sort Items, Place & View Orders, and Optimized Route Display
- Covered linked lists, sorting algorithms, graphs, and priority queues

Quiz Application

Object-Oriented Programming | Java + Java Swing

Desktop GUI application designed for quiz creation and assessment

- Built with full OOP architecture (inheritance, encapsulation, polymorphism) and Java Swing GUI
- Supports MCQs, True/False, and Fill-in-the-Blank formats with real-time evaluation
- Includes a quiz creator module allowing instructors to build and save custom quizzes

Digital Diary Application

Programming Fundamentals | Java

CLI-based diary management system with persistent file storage

- Implemented full CRUD operations: Create, Read, Update, and Delete diary entries
- Used file handling to store entries with timestamps across sessions

University Student Resource Ecosystem App

Software Requirements Engineering | Documentation

Comprehensive SRS document for a secure student resource sharing and marketplace platform

- Defined functional & non-functional requirements, use case diagrams, and event-response tables
- Modules: Ride Sharing • Hostel Search & Booking • Book Exchange • Tech Resale • User Authentication
- Incorporated Agile user stories with acceptance criteria following INVEST principles

Online Social Media Management Tool

Software Engineering | Full SDLC Documentation

End-to-end software engineering documentation project covering the complete development lifecycle

- Produced: Scope Document, SRS, SDS, Software Test Plan (STP), Testing Report, and Final Report
- Platform concept: unified dashboard to manage multiple social accounts, analytics, post scheduling, and trending keywords

JhootayShootay — E-Commerce Store

Digital Marketing | WordPress + Pantheon.io

Fully functional e-commerce store built and deployed as part of E-Commerce coursework

- Designed and deployed a complete WooCommerce store on Pantheon.io hosting platform
- Configured product listings, cart functionality, and on-page SEO optimization
- Executed social media marketing campaigns and ran Google Ads within free-tier budget

TECHNICAL SKILLS

Languages:	Java • C++ • Python • TypeScript / JavaScript
Mobile / Frontend:	React Native (Expo) • React Navigation • React
Backend:	Node.js • Express • JWT Authentication • REST APIs
Machine Learning:	Python (ML) • Data Analysis • Model Development
Databases:	Oracle SQL • MongoDB • MongoDB Atlas • Aggregation Pipelines • Index Optimization
Tools:	VS Code • Figma • Canva • WordPress • Pantheon.io • Render • EAS Build • Git • GitHub • MS Office
Concepts:	OOP • Data Structures & Algorithms • SDLC • SRS/SDS Documentation • Machine Learning
Marketing:	SEO • Google Ads • Social Media Marketing • E-Commerce Management

SOFT SKILLS

Teamwork • Problem Solving • Communication • Technical Documentation • Time Management

CERTIFICATIONS

Machine Learning Internship — Certificate of Completion*Arch Technologies | Apr – May 2026 | Cert ID: 993817*

AI-Powered Data Analysis — Certificate of Completion *Excelerate Global Remote Internship Program | 2026*

LANGUAGES

English (Professional) • **Urdu** (Native) • **Punjabi** (Native)